

6.1 Waveforms, Spectrograms, and Time–Frequency Tokens

Sound is a 1-D wave. A spectrogram is an image the Week 4 stack already knows how to read.

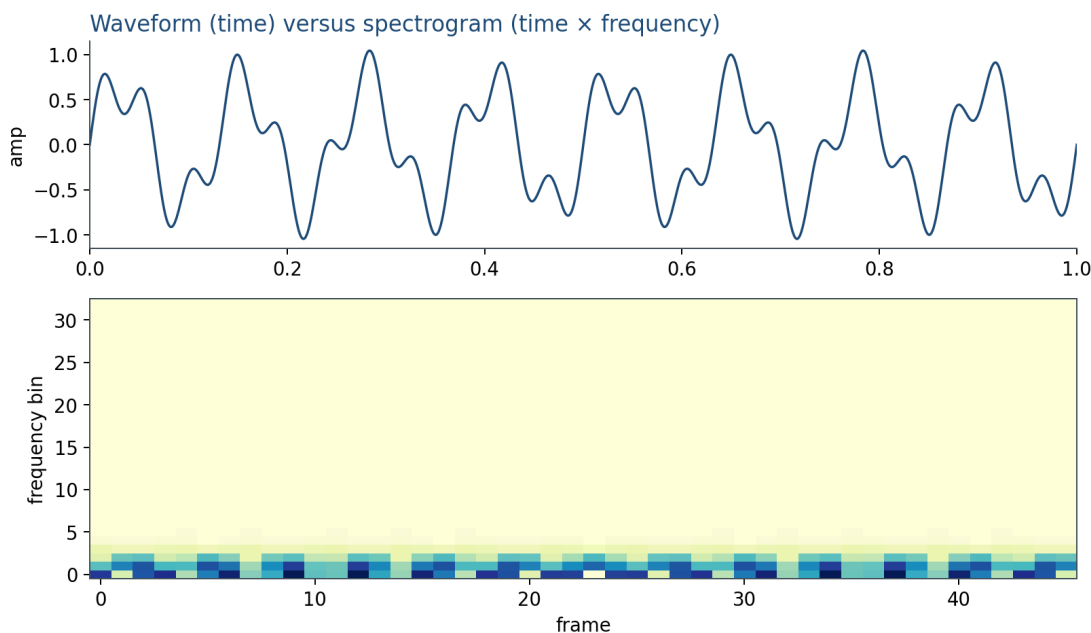
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

A waveform is a 1-D list of samples. A **spectrogram** is that list cut into short windows, each turned into a frequency snapshot. Once you have a 2-D time–frequency picture, every vision trick from Week 4 (patches, a transformer, CLIP-style towers) applies to audio.

1. Waveforms and the STFT

You record pressure $x[n]$ at a sampling rate f_s (16 kHz is a common speech default). A sine of f hertz is $x[n] = \sin(2\pi f n / f_s)$. Mixtures add; that is still one stream.

The **short-time Fourier transform (STFT)** windows the signal (Hann, hop H , FFT size N), then takes a DFT of each frame. The **magnitude spectrogram** $S(t, f) = |X(t, f)|$ is what you plot. Phase exists, but most audio encoders drop it and keep magnitudes or a **mel** compression of them (log-mel filterbank).

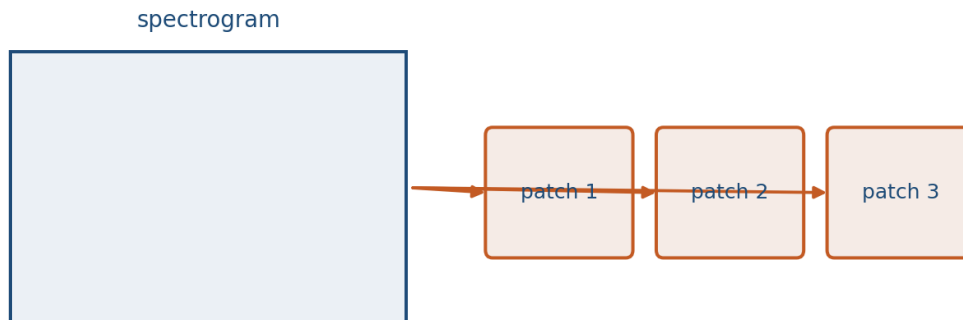


A tall band that stays put is a steady tone. A sweep that climbs is a chirp. Speech looks like stacked formants that move.

2. Spectrograms as images, then as tokens

Treat S as a single-channel image. A CNN can ingest it. A ViT can **patch** it: tiles of time $\times$ frequency, flatten, linear map to width d , add positions. Lab 6 does that cut by hand.

Same trick as ViT: cut time-frequency patches, then embed



Time is not quite space. Adjacent frequency bins are related (harmonics); adjacent frames are related (smooth speech). Positions should know which axis is which. Still, "patch then attend" is the default modern front end.

3. What this is not

A spectrogram is not a transcript, a speaker id, or a music tag. Those are **tasks** on top of the tokens. Whisper (note **6.2**) will decode text from this picture. CLAP will embed the picture next to a caption. If your project uses audio, write f_s , window, hop, and whether you used mel or raw STFT. Those choices change the tokens.

Log-mel is a compressive front end: fewer bands than a linear STFT, spaced the way hearing is. It is still a spectrogram. Do not skip the plot; if the 440 Hz band is missing, the rest of the pipeline is theater.

4. Practice

1. You double the hop length and keep N fixed. What happens to the number of time frames, and to time resolution?
2. Why is a spectrogram a better input to a transformer than the raw waveform at 16 kHz?
3. Count patch tokens for a mel grid of 80×400 with 16×16 tiles (ignore remainder).